Predictive Modeling and Risk Scoring for Bank Customer Churn

A Data-Driven Analysis of Churn Drivers, Risk Tiers, and Retention Scenarios

Research Paper
Dataset: 10,000 customer records · 3 countries · 2,037 churned customers
October 2026

Prepared from the European Bank customer dataset (“European_Bank.csv”) and an accompanying Python (scikit-learn) modelling pipeline.

Abstract

This paper presents a quantitative analysis of customer churn at a European retail bank, covering 10,000 customer records from France, Germany and Spain, of whom 2,037 (20.4%) left the bank. The study builds a predictive early-warning system that assigns every customer a churn probability and a risk tier. Four behavioural features are engineered and five classifiers (logistic regression, decision tree, random forest, gradient boosting and XGBoost) are compared using a stratified 80/20 split and five-fold cross-validation. Gradient boosting performs best, with a test ROC-AUC of 0.872 (cross-validated 0.864). At a tuned threshold of 0.32 it identifies 63.6% of churners with 64.9% precision, and customers scored High Risk churn at 88.1% versus 9.0% for Low Risk. Feature importance, SHAP values and partial dependence show that customer age and the number of products held are the dominant drivers, followed by German residence, gender and account activity. A what-if simulation suggests that re-activating inactive members and moving single-product customers to a second product could materially lower the predicted churn of the highest-risk cohort, although these are model-based associations and not proven causal effects. The paper also documents the limitations of a single cross-sectional snapshot and offers recommendations for retention strategy and data collection.

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1. Introduction

Retail banks depend on long customer relationships. Each departing customer takes with them future interest income, fee revenue and cross-sell potential, and replacing that customer is usually more expensive than keeping them. Traditional churn analysis is descriptive: it explains why customers left after the fact. Modern retention work needs the reverse, an early and quantified view of which current customers are most likely to leave, so that offers and outreach can be targeted rather than broad.

This paper analyses a customer-level dataset from a European retail bank to build such a view. It trains and compares several machine-learning classifiers, converts the best model into a probability-based risk score, and uses explainability tools to show which customer characteristics drive the score. The analysis was run on the supplied dataset with a Python pipeline that keeps all preprocessing inside a leakage-safe scikit-learn workflow.

2. Banking Context and Problem Statement

Banks hold rich customer data but often lack an accurate churn model, a quantitative risk score and a clear explanation of what drives churn. As a result, retention actions tend to be reactive rather than proactive, broad rather than targeted, and costly rather than efficient. The project therefore sets the objectives summarised below.

Item	Detail
Business problem	Identify customers likely to churn before they leave, and explain why
Target variable	Exited (1 = churned, 0 = retained); about 20.4% of customers are churners
Primary objectives	Predict churn accurately; produce a 0-1 churn probability; identify key churn drivers
Secondary objectives	Reduce false alarms; keep the model interpretable; support what-if risk analysis
Prediction output	Churn probability, binary churn flag (threshold-based) and a Low / Medium / High risk tier
Markets covered	France, Spain and Germany

Source: project brief and dataset documentation.

3. Data and Methodology

The analysis uses 10,000 customer-level records, each describing credit score, geography, gender, age, tenure, balance, number of products, credit-card ownership, activity status, estimated salary and the churn outcome. The supplied CSV also contained about 40 pre-computed dashboard columns (risk scores, flags, categories and churn-rate aggregates). These were excluded because several are derived from the target and would leak the answer. The modelling pipeline performed the following steps:

- Loaded the dataset and validated row and column counts, missing values and duplicate customer IDs (none were found).
- Dropped the non-informative identifiers CustomerId and Surname.
- One-hot encoded Geography and Gender and standardised numeric features inside a pipeline fitted only on training data.
- Engineered four features: Balance-to-Salary ratio, Product Density (products divided by tenure plus one), Engagement-Product interaction (activity x products) and Age-Tenure interaction.
- Split the data 80/20 with stratification on the target (seed 42): 8,000 training and 2,000 test customers, churn rates 20.4% and 20.3%.
- Compared five models with stratified five-fold cross-validation on the training set: logistic regression (baseline), decision tree, random forest, gradient boosting and XGBoost.
- Tuned the decision threshold on out-of-fold training predictions, then evaluated once on the test set using accuracy, precision, recall, F1, ROC-AUC, Brier score and calibration.

- Explained the selected model with built-in and permutation feature importance, SHAP values and partial dependence plots.

All figures in this paper are computed directly from the supplied dataset; no values have been estimated or supplemented from outside sources.

4. Key Performance Indicators

KPI	Value
Total customers	10,000
Customers who churned	2,037
Overall churn rate	20.4%
Average customer age	38.9 years
Average tenure	5.0 years
Average account balance	76,486
Average estimated salary	100,090
Countries served	France, Germany, Spain
Models compared	5
Selected model	Gradient Boosting
Test ROC-AUC	0.872
Cross-validated ROC-AUC	0.864 +/- 0.008
Precision / recall at tuned threshold	64.9% / 63.6%
High Risk tier: share of customers / observed churn	9.2% / 88.1%

Balance and salary are in the currency units recorded in the source file.

5. Findings

5.1 Churn by Geography and Gender

The class balance is moderately skewed: 7,963 customers were retained and 2,037 churned, so a model that predicted "retained" for everyone would reach 79.6% accuracy while finding no churners. Accuracy alone is therefore not a sufficient yardstick. Germany stands apart: German customers churn at 32.4%, twice the rate in France (16.2%) and Spain (16.7%). Women churn at 25.1% versus 16.5% for men. Credit-card ownership (20.2% vs 20.8%) and estimated salary show no meaningful relationship with churn on their own.

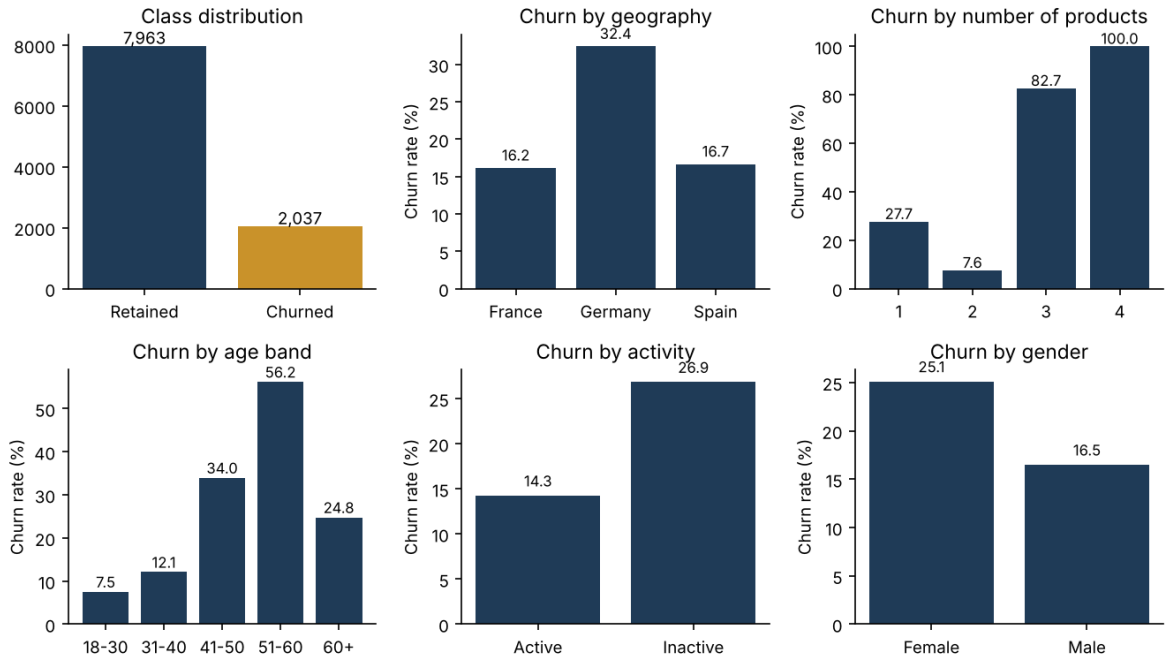


Figure 1. Class balance and churn rate by geography, number of products, age band, activity status and gender.

5.2 Churn by Age and Life Stage

Age is the strongest single signal. Churn rises from 7.5% for customers aged 18-30 and 12.1% for ages 31-40, to 34.0% for ages 41-50, and peaks at 56.2% for ages 51-60, before falling to 24.8% for customers over 60. Customers in their forties and fifties are often the highest-balance, highest-value segment, which makes this pattern commercially important.

5.3 Products and Engagement

The number of products held has a U-shaped relationship with churn. Customers with one product churn at 27.7%, those with two products at only 7.6%, but customers with three products churn at 82.7% and all 60 customers with four products churned. Account activity also matters: inactive members churn at 26.9% versus 14.3% for active members. The two-product, active customer is the most loyal profile in the dataset.

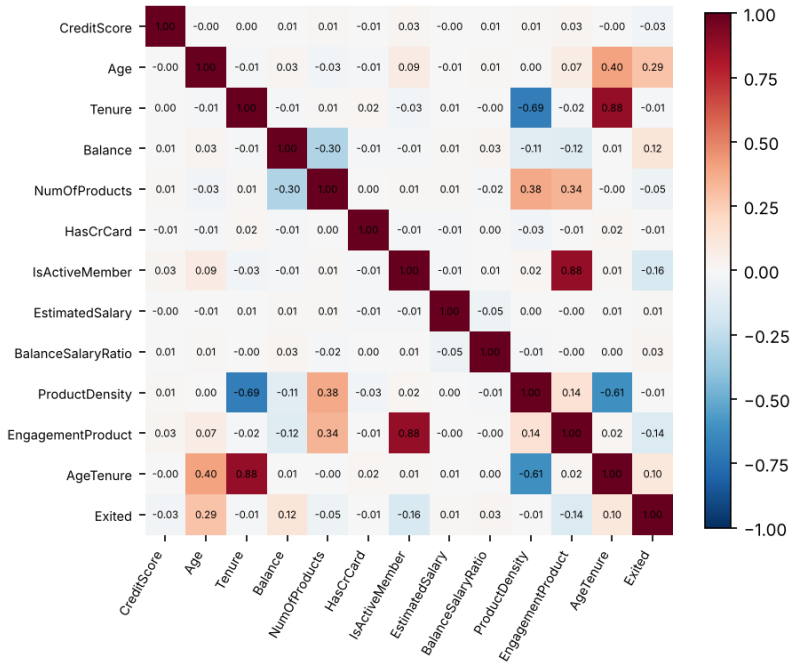


Figure 2. Correlation matrix of numeric and engineered features with the churn target. Linear correlations are modest (Age 0.29, IsActiveMember -0.16), which points toward non-linear models.

5.4 Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV AUC (5-fold)
Logistic Regression	0.710	0.383	0.698	0.494	0.776	0.766 +/- 0.021
Decision Tree	0.762	0.451	0.786	0.574	0.841	0.832 +/- 0.008
Random Forest	0.808	0.521	0.730	0.608	0.865	0.857 +/- 0.011
Gradient Boosting	0.870	0.793	0.489	0.605	0.872	0.864 +/- 0.008
XGBoost	0.870	0.787	0.491	0.605	0.865	0.863 +/- 0.010

Table 1. Test performance at the default 0.50 threshold; the highlighted row is the selected model.

Tree ensembles clearly beat the linear baseline (AUC 0.776), because churn depends on non-linear effects such as the U-shaped product curve. Gradient boosting, XGBoost and random forest are statistically close (cross-validated AUC 0.864, 0.863 and 0.857, fold standard deviation about 0.01), so their ranking should not be over-read. Gradient boosting was selected because it led on both test and cross-validated AUC and had the best-calibrated probabilities (Brier score 0.098 versus 0.139 for random forest).

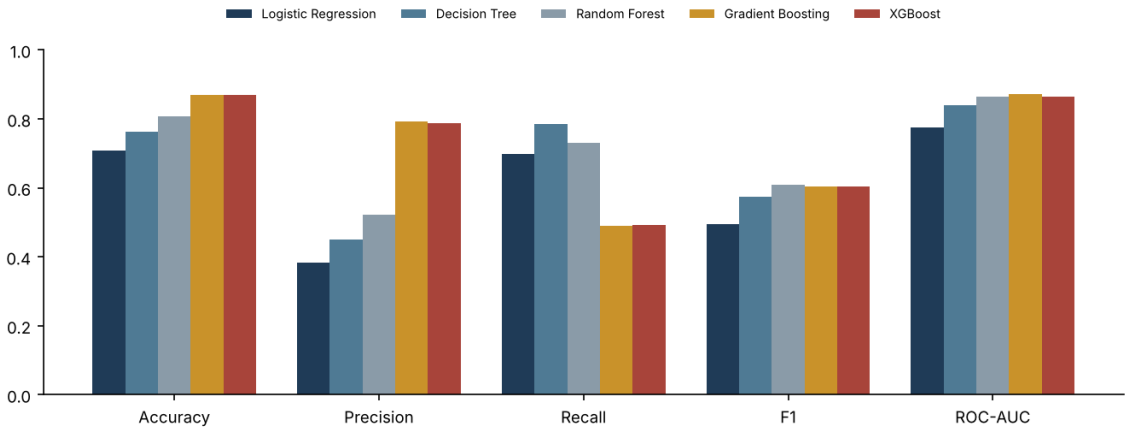


Figure 3. Test-set metrics for all five models at the 0.50 threshold.

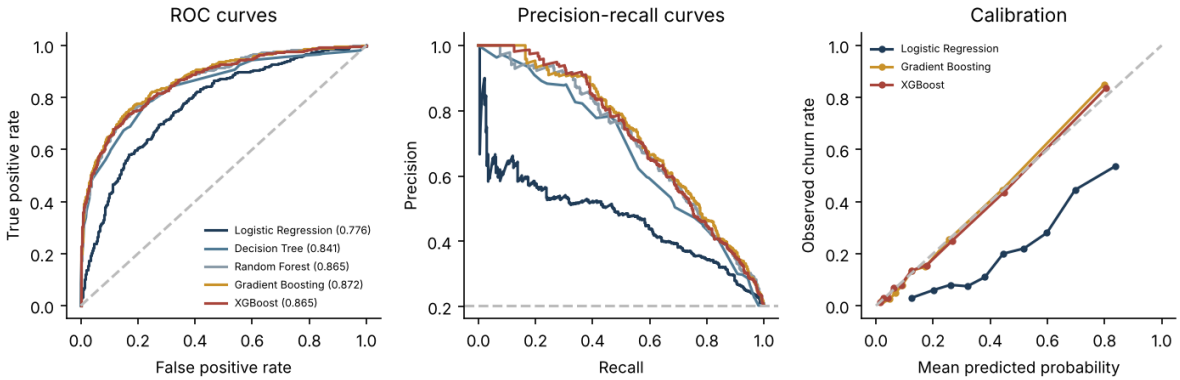


Figure 4. ROC curves, precision-recall curves and calibration curves on the test set. The boosted models track the diagonal in the calibration panel, so their scores can be read as probabilities.

An ablation retraining logistic regression and gradient boosting on the original attributes alone gave test AUC of 0.777 and 0.873, against 0.776 and 0.872 with the engineered features. The engineered features therefore added no measurable predictive power, since tree ensembles already learn these interactions from the raw columns; they are retained for interpretability and business reporting.

5.5 Decision Threshold

Operating point	Threshold	Precision	Recall	F1	Flagged / churners caught
Default	0.50	79.3%	48.9%	0.605	251 flagged / 199 of 407
Precision-oriented	0.40	71.6%	56.3%	0.630	320 flagged / 229 of 407
F1-optimised (tuned)	0.32	64.9%	63.6%	0.643	399 flagged / 259 of 407

Table 2. Effect of the decision threshold on the 2,000-customer test set (407 churners).

Lowering the threshold from 0.50 to 0.32 raises recall from 48.9% to 63.6% (60 more churners found) at the cost of 88 additional false alarms. For a low-cost action such as a phone call or in-app offer, the tuned threshold is preferable; for expensive incentives such as rate concessions, a threshold of 0.40-0.50 protects precision.

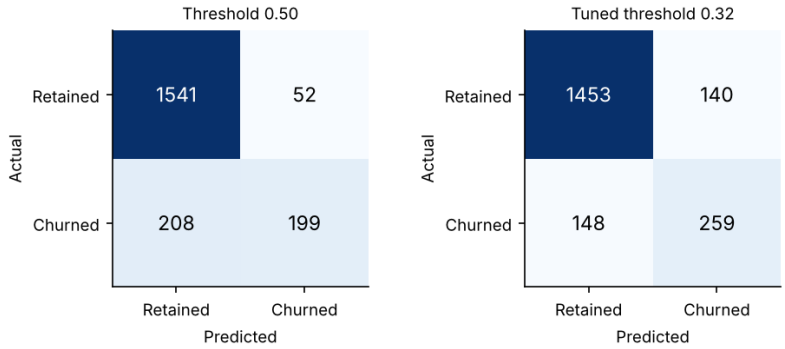


Figure 5. Confusion matrices for the selected model at the default and tuned thresholds.

5.6 Risk Tiers and Targeting Efficiency

Risk tier	Customers	Share of base	Observed churn rate	Share of all churners captured
Low (<30%)	1,568	78.4%	9.0%	34.6%
Medium (30-60%)	247	12.4%	41.7%	25.3%
High (>=60%)	185	9.2%	88.1%	40.0%

Table 3. Observed churn by predicted risk tier on the held-out test set.

The three tiers separate customers sharply: observed churn is 9.0%, 41.7% and 88.1% for Low, Medium and High risk. A team contacting only the top 10% of scored customers would reach a group in which 85.0% actually churn (4.2x the base rate) and would touch 41.8% of all churners; the top 20% captures 63.6% and the top 30% captures 76.2%. About a third of churners (35%) still sit in the Low tier because it is so large (78% of customers); no model built on these attributes is expected to find every churner.

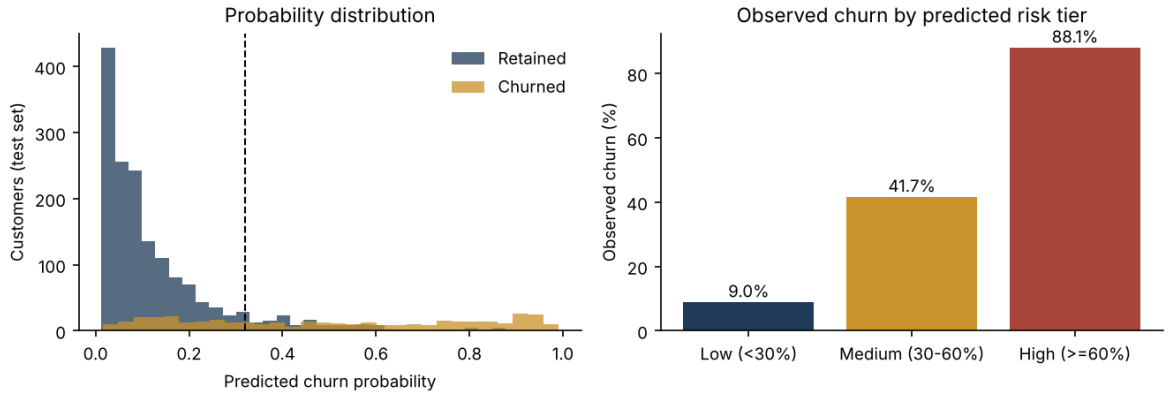


Figure 6. Distribution of predicted churn probabilities (dashed line = tuned threshold) and observed churn by predicted risk tier.

5.7 Churn Drivers and Explainability

Built-in and permutation importance agree on the top two drivers: Age (importance 0.37; permutation AUC drop 0.122) and NumOfProducts (0.27; 0.107), which together carry roughly 64% of impurity importance. The second tier contains the engineered EngagementProduct feature, Balance, IsActiveMember and German residence. CreditScore, Tenure, HasCrCard and EstimatedSalary contribute little.

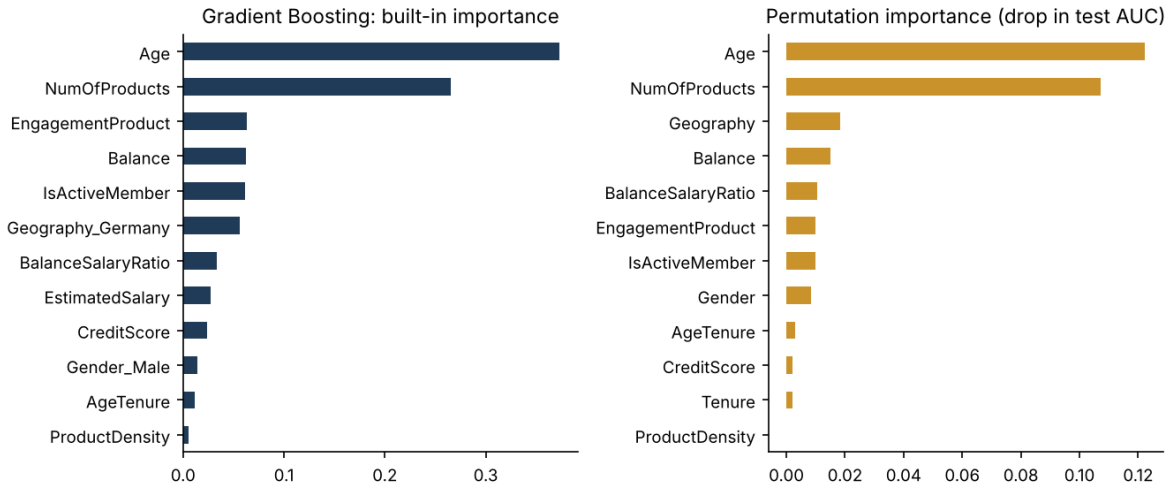


Figure 7. Gradient boosting built-in importance (left) and permutation importance as the drop in test AUC (right).

SHAP analysis confirms the direction of each effect. Mean absolute SHAP ranks NumOfProducts (0.74) and Age (0.74) far ahead of Geography_Germany (0.25), Gender_Male (0.21), EngagementProduct (0.18) and IsActiveMember (0.17). Customers with three or more products receive large positive SHAP values, two-product customers are pushed toward retention, higher age raises risk, and being German, female or inactive each adds a smaller but consistent push toward churn.

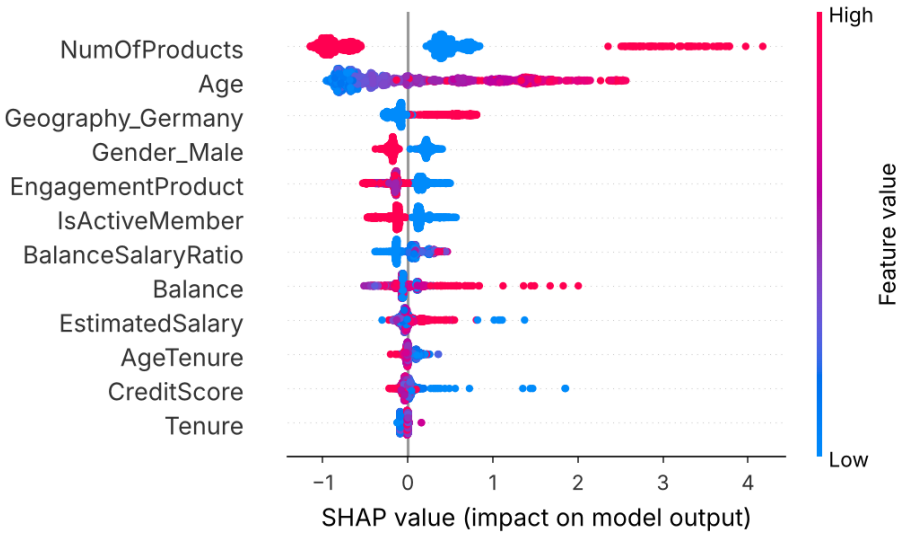


Figure 8. SHAP summary plot on the test set. Colour shows the feature value; horizontal position shows its effect on the model log-odds of churn.

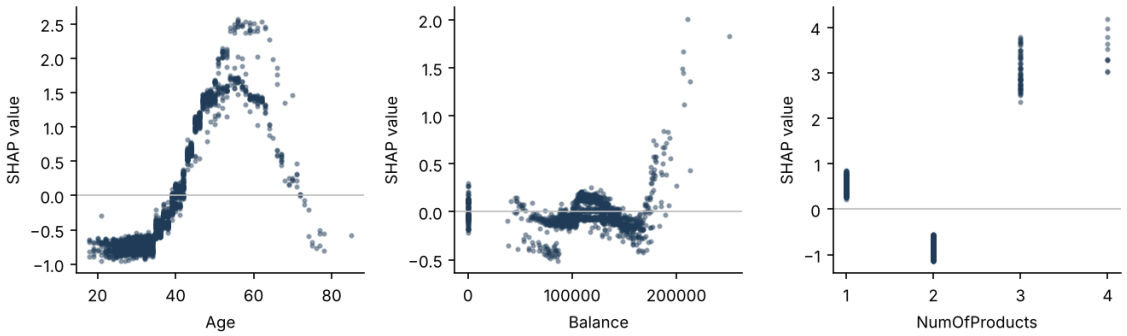


Figure 9. SHAP dependence for Age, Balance and NumOfProducts in original units.

Partial dependence shows predicted churn flat near 10% up to age 35, climbing to about 50% by the mid-fifties. For products it falls from about 27% at one product to about 12% at two, then jumps to about 75% at three. Balance and credit score move the average prediction by only about 3 and 2 percentage points, so they are weak levers on their own.

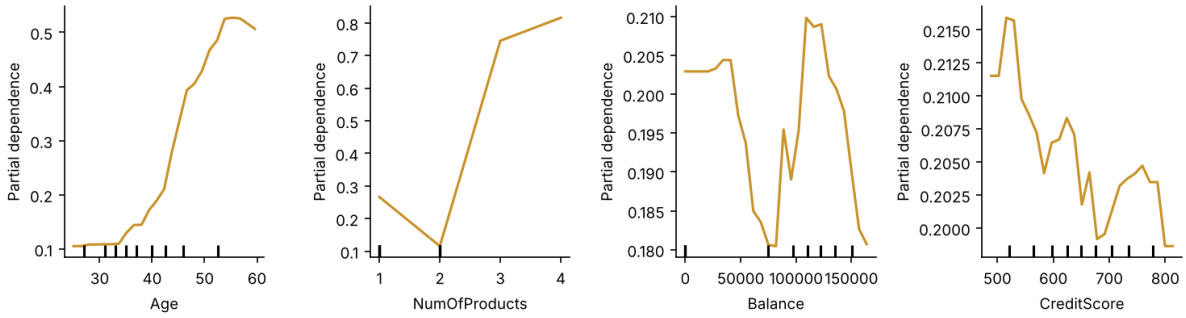


Figure 10. Partial dependence of predicted churn probability on Age, NumOfProducts, Balance and CreditScore.

As an interpretability benchmark, the logistic regression shows that each one-standard-deviation increase in Age multiplies the odds of churn by 2.27, being German (versus French) by 2.27, being an active member by 0.53 and being male by 0.58. These agree in direction with SHAP. The linear model assigns NumOfProducts an odds ratio of 0.87, hiding the U-shape, which is the main reason it trails the tree ensembles.

5.8 What-If Scenario Analysis

To show how a customer's risk would respond to changed behaviour, several scenarios were applied to the 185 High Risk test customers (mean predicted churn 81.5%), with all derived features and probabilities recomputed for each.

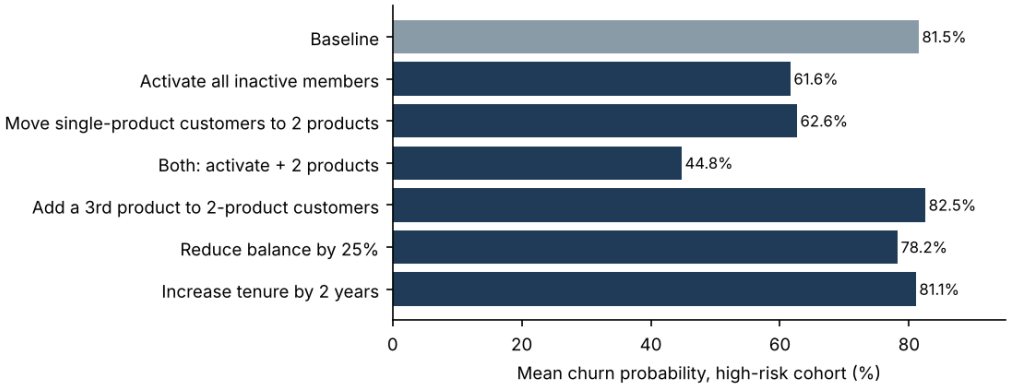


Figure 11. Mean predicted churn probability of the High Risk cohort under each scenario.

Re-activating every inactive member lowers the cohort's mean predicted churn from 81.5% to 61.6%; moving single-product customers to a second product lowers it to 62.6%; doing both reduces it to 44.8%, the largest effect tested. Adding a third product to two-product customers slightly *raises* risk (82.5%), and reducing balance or extending tenure has almost no effect. For an illustrative 45-year-old German male with two products and an inactive account, the model gives a churn probability of 36.0%; re-activating the account lowers it to 20.5%, while a move to three products raises it to 90.0%.

These simulations show how the *model* responds, based on correlations in historical data. They do not prove that nudging a customer to become active or buy a second product will cause lower churn, so they should guide a controlled pilot rather than be read as guaranteed savings.

6. Data Quality and Limitations

The dataset is clean in a mechanical sense: there are no missing values and no duplicate customer IDs. Its limitations are structural. The file is a single cross-sectional snapshot (all records stamped 2025), so behaviour *before* churn, such as transaction volumes, complaints and log-in trends, cannot be observed; a time-stamped panel would likely improve recall materially. The supplied file also mixed raw columns with dashboard columns derived from the target, which had to be excluded to avoid leakage.

Other limitations include: test metrics come from 2,000 customers (407 churners), so differences of about 0.01 AUC between the top models are within noise; the three- and four-product groups are small (266 and 60 customers) and give unstable estimates; the what-if scenarios are associational; and gender and geography are predictive, so a bank should review the lawfulness and fairness of using them in automated decisions and test for disparate impact before operational use. The data appear to be a widely used public benchmark of European retail customers, so deployment would require the bank's own current data and a back-test on a later period.

7. Discussion and Recommendations

- **Run a tiered retention programme.** Contact High Risk customers (about 9% of the base, 88% observed churn) with personal outreach, and Medium Risk customers (about 12%, 42% churn) with lower-cost digital offers. Leave the Low tier on standard service.
- **Build activity before it is lost.** Inactive members churn at almost twice the rate of active ones. Trigger dormancy alerts and re-engagement journeys such as app log-in nudges, salary-credit incentives and card-usage rewards.
- **Aim for the second product.** Two-product customers churn at only 7.6%, so a well-timed second product for the 5,084 single-product customers is the most promising cross-sell lever. Avoid pushing customers beyond two products until the 82.7% and 100% churn at three and four products is understood; it may reflect mis-selling, fee complexity or a data artefact.
- **Investigate Germany.** German customers churn at twice the rate of French and Spanish customers, which points to a market-level issue such as competition, pricing, service quality or product fit.
- **Give customers aged 41-60 tailored attention.** This group (churn 34% to 56%) is often high-value; consider wealth, retirement and mortgage-review conversations.
- **Use the threshold as a business dial.** Align the cut-off with the unit cost of each retention action and re-tune it quarterly against actual campaign response.
- **Monitor, retrain and deploy.** Track calibration and tier-level churn monthly and retrain at least annually. The saved pipeline can be served through a Streamlit application with a customer risk calculator, probability-distribution view, feature-importance dashboard and what-if simulator.

8. Conclusion

This analysis of 10,000 bank customers delivers a working churn intelligence pipeline: leakage-free data preparation, five benchmarked models, a well-calibrated gradient-boosting risk scorer (test AUC 0.872) with three risk tiers, and a multi-method explanation of its drivers. The score concentrates churn effectively: the 9% of customers flagged High Risk churn at 88%, and the top 20% of ranked customers contain 63.6% of all churners.

One finding refines the original framing of the project. Engagement and product use do matter, but age and product count are the two largest drivers, so a purely behavioural view would miss the life-stage effect. The most actionable levers are still the behavioural ones, because a bank can influence activity and product holdings but not age. The clearest next steps are a controlled retention pilot to measure causal uplift, enrichment with transactional data, a fairness review and deployment of the scoring application.

References

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